

California Housing Price Prediction

Task 1: Baseline Linear Regression Model Evaluation

Executive Summary

This technical report details the development and evaluation of a baseline **Linear Regression model** to predict California house prices using the California Housing dataset. Using features such as geographical coordinates, housing age, rooms, population, and median income, the model establishes a mathematical baseline for pricing houses. The model is evaluated on a test set (20%) using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2). This baseline will serve as a comparison benchmark for future complex modeling efforts (e.g., Random Forests, XGBoost).

Exploratory Data Analysis (EDA)

The California Housing dataset contains **20,640 records**. During the initial EDA phase, we analyzed the distribution of the target variable (median house value) and inspected feature correlations. We observed that the median house value has a significant cap at \$500,001 (creating a spike at the far right of the distribution). Additionally, we found a high correlation between median income and the median house value, highlighting income as the strongest predictor of housing prices.

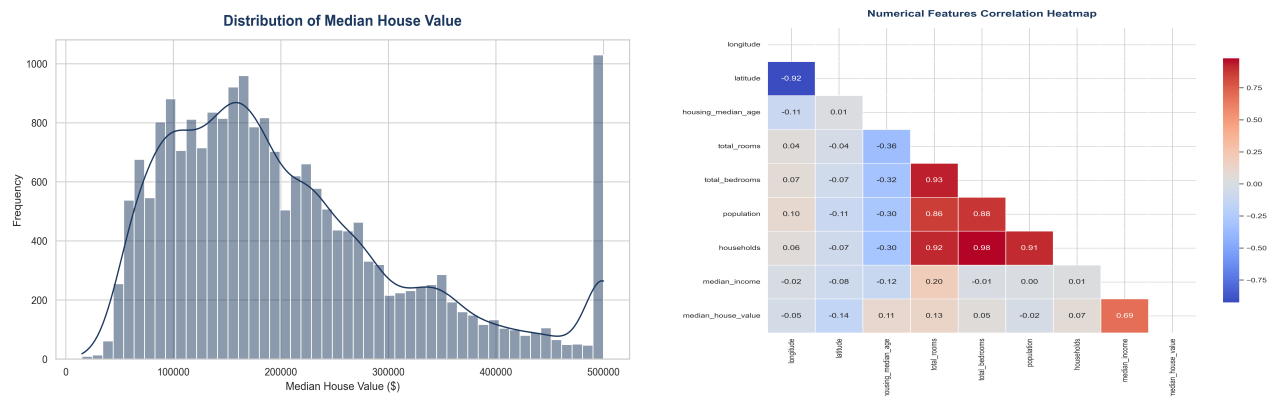


Figure 1: Distribution of Median House Value (left) and Correlation Matrix Heatmap (right)

Data Preprocessing & Pipeline Construction

To ensure robust evaluation and prevent data leakage, a strict preprocessing pipeline was constructed using scikit-learn. The pipeline automates the following steps:

- **Imputation:** The `total_bedrooms` column contained 207 missing values, which were imputed with the median of the training partition.
- **Standardization:** Numerical variables were scaled to have zero mean and unit variance using `StandardScaler`, ensuring coefficients are comparable and numeric values are scale-independent.
- **Categorical Encoding:** The `ocean_proximity` attribute was converted to binary indicator variables using `OneHotEncoder` to allow mathematical processing.
- **Split:** A standard 80/20 train/test split was performed with `random_state=42`.

Model Training & Evaluation Metrics

The model was trained on the training partition (16,512 samples) and evaluated on the test partition (4,128 samples). The evaluation metrics indicating baseline performance are summarized below:

Performance Metric	Value	Interpretation
Mean Absolute Error (MAE)	\$50,670.49	On average, the predictions deviate from the true value by this amount.
Root Mean Squared Error (RMSE)	\$70,059.19	Penalizes larger errors. Reflects the standard deviation of residuals.
Coefficient of Determination (R ²)	0.6254	The model features explain approximately 62.5% of the target's variance.

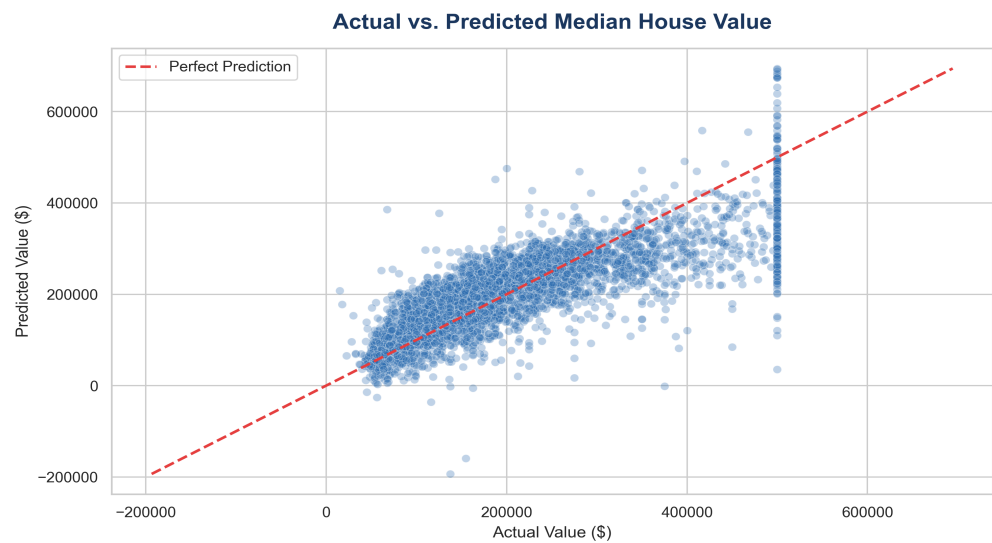


Figure 2: Actual vs. Predicted values. Dashed line indicates perfect predictions ($y = x$).

Residuals Analysis

Analyzing the distribution of residuals (errors) is critical to verify the assumptions of linear regression: linearity, homoscedasticity, and normality of errors. The residuals analysis is displayed below.

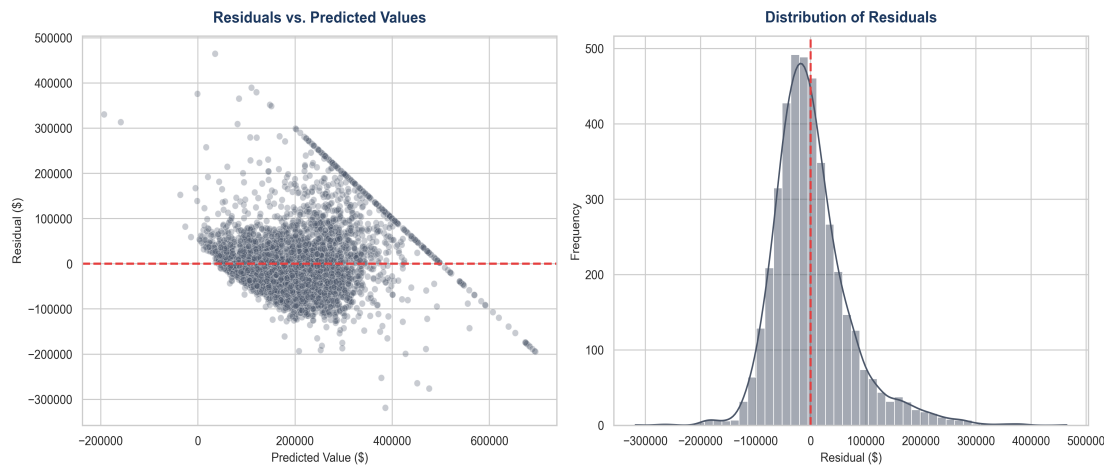


Figure 3: Residuals plot showing errors vs. predictions (left) and the distribution of errors (right).

Interpretation of Key Model Coefficients

The model intercept is **\$238,826.36**. The table below lists the coefficients associated with features after numerical standardization, indicating their relative importance and impact:

Feature	Coefficient	Direction
ocean_proximity_ISLAND	\$117,198.49	Positive (Increases Price)
median_income	\$75,167.77	Positive (Increases Price)
ocean_proximity_INLAND	-\$58,713.24	Negative (Decreases Price)
latitude	-\$54,415.70	Negative (Decreases Price)
longitude	-\$53,826.65	Negative (Decreases Price)
population	-\$43,403.43	Negative (Decreases Price)

Key Insights & Actionable Improvement Ideas

- Underlying Limitations of Linear Regression:** A linear model assumes linear relationships and additive effects, which fail to capture the complex, non-linear geographical interactions in housing data (e.g. proximity to tech hubs or specific coastline boundaries).
- Improvement via Advanced Algorithms:** Transitioning to tree-based ensemble methods (e.g., Random Forests, Gradient Boosted Trees) will capture non-linear feature interactions and handle spatial coordinates

(latitude and longitude) more effectively.

3. **Feature Engineering Ratios:** Introducing engineered attributes such as *rooms per household*, *bedrooms per room*, or *population per household* provides stronger indicators of house size and density compared to raw aggregate figures.
4. **Spatial Geofencing:** Adding distances to major employment centers (San Francisco, Los Angeles) or clustering geolocations will improve prediction accuracy dramatically.